

Shichang Zhang

CONTACT INFORMATION	Address: 300 E Lemon St. BAC 552, Tempe, AZ 85287 Email: shichang.zhang@asu.edu Website: https://shichangzh.github.io/	
WORK EXPERIENCE	Arizona State University <i>Assistant Professor of Information Systems, W.P. Carey School of Business</i>	Tempe, AZ July 2026 - Present
	Harvard University <i>Postdoctoral Fellow, Harvard Business School AI Institute</i>	Cambridge, MA Aug. 2024 - June 2026
EDUCATION	University of California, Los Angeles <i>Ph.D. in Computer Science</i>	Los Angeles, CA June 2024
	Stanford University <i>M.S. in Statistics</i>	Stanford, CA Apr. 2019
	University of California, Berkeley <i>B.A. in Statistics</i> Honors: Honors in Statistics, High Distinction	Berkeley, CA May 2017
RESEARCH INTERESTS	Explainable AI, Trustworthy AI, Data Attribution, Mechanistic Interpretability, Large Language Models, Graph Data Mining	
HONORS AND AWARDS	NENLP Outstanding Paper KDD Outstanding Reviewer (Top 10%, twice, in Aug and Feb) Amazon PhD Fellowship J.P.Morgan Chase AI PhD Fellowship KDD Excellence in Reviewing (Top 30 of 1551) Snap Research Fellowship Honorable Mention ICML Top Reviewer (Top 10%) UCLA Graduate Division Fellowship	2025 2025 2023 2023 2023 2022 2022 2021
PUBLICATIONS	Refereed Publications: 1. Who Gets Credit or Blame? Attributing Accountability in Modern AI Systems Shichang Zhang , Hongzhe Du, Jiaqi W. Ma, Himabindu Lakkaraju International Conference on Machine Learning (ICML), 2026 2. Explainability Research Must Prioritize Foundations over Ad-hoc Methods Michal Moshkovitz, Suraj Srinivas, Lesia Semenova, Nave Frost, Cyrus Rashtchian, Valentyn Boreiko, Shichang Zhang , Himabindu Lakkaraju, Cynthia Rudin, Jennifer Wortman Vaughan International Conference on Machine Learning (ICML), 2026	

3. Efficient Ensembles Improve Training Data Attribution
Junwei Deng*, Ting-Wei Li*, **Shichang Zhang**, Jiaqi Ma (*equal contribution)
Transactions on Machine Learning Research (**TMLR**), 2026
4. A Survey on Graph Neural Network Acceleration: Algorithms, Systems, and Customized Hardware
Shichang Zhang, Atefeh Sohrabizadeh, Cheng Wan, Zijie Huang, Ziniu Hu, Yewen Wang, Yingyan (Celine) Lin, Jason Cong, Yizhou Sun
ACM Computing Survey (**CSUR**), 2026
5. How Post-Training Reshapes LLMs: A Mechanistic View on Knowledge, Truthfulness, Refusal, and Confidence
Hongzhe Du*, Weikai Li*, Min Cai, Karim Saraipour, Zimin Zhang, Himabindu Lakkaraju, Yizhou Sun, **Shichang Zhang** (*equal contribution)
Conference on Language Modeling (**COLM**), (**NENLP Outstanding Paper**), 2025
6. Automated Molecular Concept Generation and Labeling with Large Language Models
Zimin Zhang*, Qianli Wu*, Botao Xia*, Fang Sun, Ziniu Hu, Yizhou Sun, **Shichang Zhang** (*equal contribution)
International Conference on Computational Linguistics (**COLING**), 2025
7. An Explainable AI Approach using Graph Learning to Predict ICU Length of Stay
Tianjian Guo, Indranil Bardhan, Ying Ding, **Shichang Zhang**
Information Systems Research (**ISR**), 2024
8. Motif-driven Contrastive Learning of Graph Representations
Shichang Zhang*, Ziniu Hu*, Arjun Subramonian, Yizhou Sun (*equal contribution)
IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2024
9. Predicting and Interpreting Energy Barriers of Metallic Glasses with Graph Neural Networks
Haoyu Li*, **Shichang Zhang***, Longwen Tang, Yizhou Sun (*equal contribution)
International Conference on Machine Learning (**ICML**), 2024
10. PaGE-Link: Graph Neural Network Explanation for Heterogeneous Link Prediction
Shichang Zhang, Jiani Zhang, Xiang Song, Soji Adeshina, Da Zheng, Christos Faloutsos, Yizhou Sun
The Web Conference (**WWW**), 2023
11. GStarX: Explaining Graph Neural Networks with Structure-Aware Cooperative Games
Shichang Zhang, Yozen Liu, Neil Shah, Yizhou Sun
Advances in Neural Information Processing Systems (**NeurIPS**), 2022
12. Graph-less Neural Networks: Teaching Old MLPs New Tricks via Distillation
Shichang Zhang, Yozen Liu, Yizhou Sun, Neil Shah
International Conference on Learning Representations (**ICLR**), 2022
13. Weak Models Can be Good Teachers: A Case Study on Link Prediction with MLPs
Zongyue Qin, **Shichang Zhang**, Mingxuan Ju, Tong Zhao, Neil Shah, Yizhou Sun
Learning on Graphs Conference (**LoG**), 2025
14. FUSE: Measure-Theoretic Compact Fuzzy Set Representation for Taxonomy Expansion
Fred Xu, Song Jiang, Zijie Huang, Xiao Luo, **Shichang Zhang**, Yuanzhou Chen, Yizhou Sun
Findings of the Association for Computational Linguistics (**ACL Findings**), 2024
15. SciBench Evaluating College-Level Scientific Problem-Solving Abilities of Large Language Models
Xiaoxuan Wang*, Ziniu Hu*, Pan Lu*, Yanqiao Zhu*, Jieyu Zhang, Satyen Subramaniam, Arjun R Loomba, **Shichang Zhang**, Yizhou Sun, Wei Wang (*equal contribution)
International Conference on Machine Learning (**ICML**), 2024

16. Laplacian Score Benefit Adaptive Filter Selection for Graph Neural Networks
Yewen Wang, **Shichang Zhang**, Junghoo Cho, Yizhou Sun
SIAM International Conference on Data Mining (**SDM**), 2024
17. Linkless Link Prediction via Relational Distillation
Zhichun Guo, William Shiao, **Shichang Zhang**, Yozen Liu, Nitesh Chawla, Neil Shah, Tong Zhao
International Conference on Machine Learning (**ICML**), 2023
18. Graph Condensation for Graph Neural Networks
Wei Jin, Lingxiao Zhao, **Shichang Zhang**, Yozen Liu, Jiliang Tang, Neil Shah.
International Conference on Learning Representations (**ICLR**), 2022

Preprints:

1. User Persona Subspaces Modulate Refusal Behavior in Language Models
Yan Zhou, **Shichang Zhang**, Zidi Xiong, Himabindu Lakkaraju
(MechInterp@ICML), 2026
2. How Faithful Is Trajectory-Based Data Attribution? Error Sources, Remedies, and Practical Guidelines
Junwei Deng, Pingbang Hu, Suliang Jin, Hao Lu, Jiachen T. Wang, **Shichang Zhang**, Jiaqi W. Ma
(Under Review), 2026
3. Towards Unified Attribution in Explainable AI, Data-Centric AI, and Mechanistic Interpretability
Shichang Zhang, Tessa Han, Usha Bhalla, Himabindu Lakkaraju
(Under Review), 2025
4. Generalized Group Data Attribution
Dan Ley, **Shichang Zhang**, Suraj Srinivas, Gili Rusak, Himabindu Lakkaraju
(Under Review), 2024
5. Computational Copyright: Towards A Royalty Model for Music Generative AI
Junwei Deng, Xirui Jiang, Shiyuan Zhang, **Shichang Zhang**, Himabindu Lakkaraju, Ruijiang Gao, Chris Donahue, Jiaqi W. Ma
(Under Review), 2025
6. From Indirect Object Identification to Syllogisms: Exploring Binary Mechanisms in Transformer Circuits
Karim Saraipour, **Shichang Zhang**
(MechInterp@NeurIPS), 2025
7. Hierarchical Compression of Text-Rich Graphs via Large Language Models
Shichang Zhang, Da Zheng, Jiani Zhang, Qi Zhu, Xiang Song, Soji Adeshina, Christos Faloutsos, George Karypis, Yizhou Sun
(Preprint), 2024
8. Self-Control of LLM Behaviors by Compressing Suffix Gradient into Prefix Controller
Min Cai, Yuchen Zhang, **Shichang Zhang**, Fan Yin, Difan Zou, Yisong Yue, Ziniu Hu
(MechInterp@ICML), 2024
9. Parameter-Efficient Tuning Large Language Models for Graph Representation Learning
Qi Zhu, Da Zheng, Xiang Song, **Shichang Zhang**, Bowen Jin, Yizhou Sun, George Karypis
(Preprint), 2024

MEDIA COVERAGE	Unifying AI Attribution: A New Frontier in Understanding Complex Systems <i>HBS AI Insights</i>	June 2025
	ChatGPT has entered the classroom: how LLMs could transform education <i>Nature News Feature</i>	Nov. 2023
TEACHING EXPERIENCE	Instructor , Arizona State University CIS508: Machine Learning in Business	Fall 2026
	Instructor , University of California, Los Angeles CS97: Introduction to Data Science	Summer 2024
	Teaching Assistant , University of California, Los Angeles CS145: Introduction to Data Mining CS32: Introduction to Computer Science II	Fall 2020, Fall 2021 Spring 2021
MENTORSHIP	Arjun Subramonian (UCLA Undergrad → UCLA PhD)	Mar. 2020 - Mar. 2021
	Qianli Wu (UCLA Undergrad → Amazon SDE)	Mar. 2023 - Mar. 2024
	Haoyu Li (UCLA Undergrad → UIUC PhD)	Mar. 2023 - July 2024
	Gaotang Li (UMich Undergrad → UIUC PhD)	Oct. 2023 - June 2024
	Botao Xia (UCLA Undergrad → UCLA Master)	Oct. 2023 - Aug. 2024
	Zimin Zhang (UCLA Undergrad → UIUC Master)	Oct. 2023 - Dec. 2025
	Min Cai (Shenzhen University Master → UAlberta PhD)	Nov. 2023 - Dec. 2025
	Hongzhe Du (UCLA Master)	Mar. 2024 - Dec. 2025
	Karim Saraipour (UCLA Master → Amazon SDE)	Apr. 2024 - Dec. 2025
	Weikai Li (UCLA Ph.D.)	Sept. 2024 - Present
	Dan Ley (Harvard Ph.D.)	Sept. 2024 - Dec. 2025
	Ethan Ji (UCLA Master → UCLA PhD)	June 2025 - Dec. 2025
	Terry Zhou (Harvard Master)	Sept. 2025 - Present
	Hugo Tierrablanca (Harvard Master)	Feb. 2026 - Present
	Yihan Wang (THU Undergrad)	Mar. 2026 - Present
TALKS	A Holistic Scientific Understanding for Trustworthy AI Arizona State University Mohamed bin Zayed University of Artificial Intelligence University of Tennessee Knoxville Pennsylvania State University University of Massachusetts Boston	Jan - Feb 2026
	Explain AI Models: Methods and Opportunities in Explainable AI, Data-Centric AI, and Mechanistic Interpretability NeurIPS Tutorial	Dec 2025
	How Post-Training Reshapes LLMs New England NLP Meeting	Apr 2025
	Peering into the Mind of AI Seminar at Georgia Institute of Technology	Apr 2025

	Interpreting AI Systems Through Features, Data, and Model Components Data Mining Seminar at Emory University	Apr 2025
	Explainable AI for Graph Data and More AI4LIFE Group at Harvard	Feb 2024
	Graph Neural Network Explanation for Heterogeneous Link Prediction Amazon Trans.AI Research Talk Series International World Wide Web Conference	July 2023 May 2023
	Structure-Aware Graph Neural Network Explanation AI TIME NeurIPS Talk Series	Feb 2023
	Graph-less Neural Networks NVIDIA GNN Reading Group	May 2022
ACADEMIC SERVICE	Conference Area Chair: ICLR - International Conference on Learning Representations NeurIPS - Advances in Neural Information Processing Systems ACL ARR - Association for Computational Linguistics Rolling Review	2027 2026 2025
	Conference Reviewer/Program Committee: NeurIPS - Advances in Neural Information Processing Systems ICML - International Conference on Machine Learning ICLR - International Conference on Learning Representations KDD - ACM SIGKDD Knowledge Discovery and Data Mining AAAI - AAAI Conference on Artificial Intelligence WSDM - ACM International Web Search and Data Mining Conference CIKM - ACM Conference on Information and Knowledge Management COLM - Conference on Language Modeling ACL ARR - Association for Computational Linguistics Rolling Review SDM - SIAM International Conference on Data Mining LOG - Learning on Graphs Conference ICDM - IEEE International Conference on Data Mining	2021 - 2025 2022 - 2025 2024 - 2026 2020, 2023 - 2026 2023 - 2025 2023 - 2025 2022 - 2023 2026 2026 2024 2023 2021
	Journal Reviewer: Management Science TPAMI - IEEE Transactions on Pattern Analysis and Machine Intelligence TKDD - ACM Transactions on Knowledge Discovery from Data TKDE - IEEE Transactions on Knowledge and Data Engineering TNNLS - IEEE Transactions on Neural Networks and Learning Systems TAI - IEEE Transactions on Artificial Intelligence	
	Workshop Organizer: Workshop on Regulatable Machine Learning @ NeurIPS	2024 - 2025
	Reading Group Organizer: UCLA Data Mining Reading Group	2022 - 2024
INDUSTRY WORK EXPERIENCE	Amazon Web Service (AWS)	Santa Clara, CA

Applied Scientist Intern, Graph Machine Learning Team June 2023 - Nov. 2023

- Proposed a framework for applying LLMs to text-rich graph data with hierarchical neighborhood compression, which allows LLMs to leverage the graph structure and handle long input text features gathered in a rich neighborhood.
- The proposed method outperformed traditional graph ML models on node classification benchmarks and will be incorporated into the Amazon DGL project.

Amazon Web Service (AWS) Santa Clara, CA
Applied Scientist Intern, Graph Machine Learning Team June 2022 - Oct. 2022

- Proposed a new framework to explain GNN link prediction for recommendation on graph data, which improves user trust in the model and helps developers debug the model. Work published in WWW 2023.
- The implemented framework will be incorporated into the Amazon Neptune ML project in production.

Snap Research Los Angeles, CA
Research Intern, Computational Social Science Team June 2021 - Sept. 2021

- Proposed a cross-model distillation framework to transfer knowledge from GNNs to MLPs, which speeds up model inference by 179 times and facilitates model deployment on latency-constraint applications. Work published in ICLR 2022.
- Worked on condensing large-scale training graphs to small synthetic graphs by over 90% reduction rate while maintaining competitive model performance for GNNs trained from scratch, which significantly saves storage space and achieves efficient continual learning. Work published in ICLR 2022.

WeWork Inc. Palo Alto, CA
Data Scientist Intern, Research and Applied Science Team June 2019 - Sept. 2019

- Implemented a data processing pipeline in SQL and Python for data querying, data cleaning, and feature engineering.
- Trained a Gradient Boosted Tree model on two million customer data to predict occupancy rate for WeWork buildings and achieved 0.093 MAE on the test set.
- Presented the pricing model as a selected outstanding project to the Research and Applied Science team including the VP.

SKILLS Programming: Python (PyTorch, Hugging Face, DGL), C++, R, Java, Linux, Git
Natural Languages: Mandarin Chinese (Native), English (Proficient)